

Two-Layer Model with Marangoni Recirculation for Coffee Ring Suppression

Mathematical Theory and Numerical Implementation

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Abstract

This document extends the depth-averaged coffee ring model to include Marangoni effects through a two-layer formulation. By splitting the droplet into surface and bulk layers with different velocities, we capture the essential physics of Marangoni recirculation, which can suppress coffee ring formation. We derive the velocity profiles from lubrication theory, compute layer-averaged velocities, and formulate coupled transport equations that capture particle trapping by recirculation cells.

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1 Introduction

1.1 Motivation

The single-layer depth-averaged model successfully predicts coffee ring formation but has a fundamental limitation: it cannot capture **Marangoni recirculation**. When surface tension gradients exist (due to temperature or concentration variations), they drive surface flows that can create recirculation cells. These cells can trap particles and **suppress the coffee ring**, leading to more uniform deposits.

1.2 Marangoni Effect

The Marangoni effect arises from surface tension gradients along the liquid-gas interface:

Marangoni Stress

A gradient in surface tension γ along the interface creates a shear stress:

$$\tau_{Ma} = \frac{\partial \gamma}{\partial r} \quad (1)$$

Surface tension gradients can arise from:

- **Thermal Marangoni:** Temperature variations due to evaporative cooling
- **Solutal Marangoni:** Concentration variations of surface-active species

1.3 Two-Layer Approach

To capture Marangoni effects within a computationally tractable framework, we:

1. Derive the full height-dependent velocity profile $u_r(r, z)$
2. Split the film into two layers: **surface** ($h/2 < z < h$) and **bulk** ($0 < z < h/2$)
3. Average the velocity within each layer
4. Track particle concentration in each layer separately
5. Model exchange between layers due to diffusion and vertical advection

2 Velocity Profiles from Lubrication Theory

2.1 Governing Equations

For thin films with $h \ll R$, lubrication theory applies. The radial momentum equation reduces to:

$$\mu \frac{\partial^2 u_r}{\partial z^2} = \frac{\partial p}{\partial r} \quad (2)$$

where μ is dynamic viscosity and p is pressure.

2.2 Boundary Conditions

Location	Condition
$z = 0$ (substrate)	No-slip: $u_r = 0$
$z = h$ (free surface)	Marangoni stress: $\mu \frac{\partial u_r}{\partial z} = \frac{\partial \gamma}{\partial r}$

2.3 Decomposition of Velocity

The total velocity is the sum of capillary (pressure-driven) and Marangoni (stress-driven) components:

$$u_r(r, z) = u_r^{cap}(r, z) + u_r^{Ma}(r, z) \quad (3)$$

3 Capillary Velocity Profile

3.1 Derivation

Capillary Velocity Profile

For the capillary component, the surface boundary condition is stress-free:

$$\left. \frac{\partial u_r^{cap}}{\partial z} \right|_{z=h} = 0$$

Integrate equation (2) twice:

$$u_r^{cap} = \frac{1}{2\mu} \frac{\partial p}{\partial r} z^2 + Az + B$$

Apply boundary conditions:

- At $z = 0$: $u_r^{cap} = 0 \Rightarrow B = 0$
- At $z = h$: $\frac{\partial u_r^{cap}}{\partial z} = 0 \Rightarrow \frac{1}{\mu} \frac{\partial p}{\partial r} h + A = 0 \Rightarrow A = -\frac{h}{\mu} \frac{\partial p}{\partial r}$

Therefore:

$$u_r^{cap} = \frac{1}{2\mu} \frac{\partial p}{\partial r} (z^2 - 2hz) = -\frac{1}{2\mu} \frac{\partial p}{\partial r} z(2h - z)$$

3.2 Expressing in Terms of Depth-Averaged Velocity

The depth-averaged capillary velocity is:

$$\begin{aligned} \bar{u}_r^{cap} &= \frac{1}{h} \int_0^h u_r^{cap} dz = -\frac{1}{2\mu h} \frac{\partial p}{\partial r} \int_0^h z(2h - z) dz \\ &= -\frac{1}{2\mu h} \frac{\partial p}{\partial r} \left[hz^2 - \frac{z^3}{3} \right]_0^h = -\frac{1}{2\mu h} \frac{\partial p}{\partial r} \cdot \frac{2h^3}{3} = -\frac{h^2}{3\mu} \frac{\partial p}{\partial r} \end{aligned} \quad (4)$$

Inverting for the pressure gradient:

$$\frac{\partial p}{\partial r} = -\frac{3\mu \bar{u}_r^{cap}}{h^2} \quad (5)$$

Substituting back:

Capillary Velocity Profile

$$u_r^{cap}(r, z) = \frac{3\bar{u}_r^{cap}}{2h^2} z(2h - z) \quad (6)$$

Verification:

- At $z = 0$: $u_r^{cap} = 0 \checkmark$
- At $z = h$: $u_r^{cap} = \frac{3\bar{u}_r^{cap}}{2h^2} \cdot h \cdot h = \frac{3}{2}\bar{u}_r^{cap}$ (surface velocity)

- Average: $\frac{1}{h} \int_0^h u_r^{cap} dz = \bar{u}_r^{cap} \checkmark$

4 Marangoni Velocity Profile

4.1 Derivation

Marangoni Velocity Profile

For pure Marangoni flow (no pressure gradient), equation (2) becomes:

$$\mu \frac{\partial^2 u_r^{Ma}}{\partial z^2} = 0$$

This implies u_r^{Ma} is linear in z :

$$u_r^{Ma} = Az + B$$

Apply boundary conditions:

- At $z = 0$: No-slip $\Rightarrow B = 0$
- At $z = h$: Marangoni stress $\Rightarrow \mu A = \frac{\partial \gamma}{\partial r} \Rightarrow A = \frac{1}{\mu} \frac{\partial \gamma}{\partial r}$

Therefore:

$$u_r^{Ma} = \frac{z}{\mu} \frac{\partial \gamma}{\partial r}$$

Marangoni Velocity Profile

$$u_r^{Ma}(r, z) = \frac{z}{\mu} \frac{\partial \gamma}{\partial r} \quad (7)$$

This is a linear profile: zero at the substrate, maximum at the surface.

4.2 Surface Marangoni Velocity

Define the surface Marangoni velocity:

$$u_s^{Ma} \equiv u_r^{Ma}(z = h) = \frac{h}{\mu} \frac{\partial \gamma}{\partial r} \quad (8)$$

The depth-averaged Marangoni velocity is:

$$\bar{u}_r^{Ma} = \frac{1}{h} \int_0^h \frac{z}{\mu} \frac{\partial \gamma}{\partial r} dz = \frac{1}{\mu h} \frac{\partial \gamma}{\partial r} \cdot \frac{h^2}{2} = \frac{h}{2\mu} \frac{\partial \gamma}{\partial r} = \frac{u_s^{Ma}}{2} \quad (9)$$

5 Total Velocity Profile

5.1 Combined Profile

The total radial velocity is:

$$u_r(r, z) = \underbrace{\frac{3\bar{u}_r^{cap}}{2h^2} z(2h - z)}_{\text{Capillary (half-parabola)}} + \underbrace{\frac{u_s^{Ma}}{h} z}_{\text{Marangoni (linear)}} \quad (10)$$

5.2 Velocity at Key Heights

Location	Capillary	Marangoni
$z = 0$ (substrate)	0	0
$z = h/2$ (midplane)	$\frac{9}{8}\bar{u}_r^{cap}$	$\frac{1}{2}u_s^{Ma}$
$z = h$ (surface)	$\frac{3}{2}\bar{u}_r^{cap}$	u_s^{Ma}

6 Two-Layer Model

6.1 Layer Definition

We split the film at $z = h/2$:

- **Surface layer:** $h/2 < z < h$, thickness $h_s = h/2$
- **Bulk layer:** $0 < z < h/2$, thickness $h_b = h/2$

6.2 Surface Layer Velocity

Surface Layer Average Velocity

Capillary component:

$$\begin{aligned}
 u_s^{cap} &= \frac{2}{h} \int_{h/2}^h \frac{3\bar{u}_r^{cap}}{2h^2} z(2h - z) dz = \frac{3\bar{u}_r^{cap}}{h^3} \int_{h/2}^h (2hz - z^2) dz \\
 &= \frac{3\bar{u}_r^{cap}}{h^3} \left[hz^2 - \frac{z^3}{3} \right]_{h/2}^h
 \end{aligned} \tag{11}$$

Evaluating:

- At $z = h$: $h \cdot h^2 - \frac{h^3}{3} = h^3 - \frac{h^3}{3} = \frac{2h^3}{3}$
- At $z = h/2$: $h \cdot \frac{h^2}{4} - \frac{h^3}{24} = \frac{h^3}{4} - \frac{h^3}{24} = \frac{6h^3 - h^3}{24} = \frac{5h^3}{24}$

$$\text{Difference: } \frac{2h^3}{3} - \frac{5h^3}{24} = \frac{16h^3 - 5h^3}{24} = \frac{11h^3}{24}$$

Therefore:

$$u_s^{cap} = \frac{3\bar{u}_r^{cap}}{h^3} \cdot \frac{11h^3}{24} = \frac{11}{8}\bar{u}_r^{cap}$$

Marangoni component:

$$\begin{aligned}
 u_s^{Ma, layer} &= \frac{2}{h} \int_{h/2}^h \frac{u_s^{Ma}}{h} z dz = \frac{2u_s^{Ma}}{h^2} \left[\frac{z^2}{2} \right]_{h/2}^h \\
 &= \frac{u_s^{Ma}}{h^2} \left(h^2 - \frac{h^2}{4} \right) = \frac{3}{4}u_s^{Ma}
 \end{aligned} \tag{12}$$

Surface Layer Velocity

$$u_s = \frac{11}{8}\bar{u}_r^{cap} + \frac{3}{4}u_s^{Ma} \tag{13}$$

6.3 Bulk Layer Velocity

Bulk Layer Average Velocity

Capillary component:

$$\begin{aligned} u_b^{cap} &= \frac{2}{h} \int_0^{h/2} \frac{3\bar{u}_r^{cap}}{2h^2} z(2h - z) dz = \frac{3\bar{u}_r^{cap}}{h^3} \left[hz^2 - \frac{z^3}{3} \right]_0^{h/2} \\ &= \frac{3\bar{u}_r^{cap}}{h^3} \cdot \frac{5h^3}{24} = \frac{5}{8} \bar{u}_r^{cap} \end{aligned} \quad (14)$$

Marangoni component:

$$u_b^{Ma, layer} = \frac{2}{h} \int_0^{h/2} \frac{u_s^{Ma}}{h} z dz = \frac{2u_s^{Ma}}{h^2} \cdot \frac{h^2}{8} = \frac{1}{4} u_s^{Ma} \quad (15)$$

Bulk Layer Velocity

$$u_b = \frac{5}{8} \bar{u}_r^{cap} + \frac{1}{4} u_s^{Ma} \quad (16)$$

6.4 Verification

The average of the two layers should recover the total depth-averaged velocity:

$$\begin{aligned} \frac{u_s + u_b}{2} &= \frac{1}{2} \left(\frac{11}{8} + \frac{5}{8} \right) \bar{u}_r^{cap} + \frac{1}{2} \left(\frac{3}{4} + \frac{1}{4} \right) u_s^{Ma} \\ &= \frac{1}{2} \cdot 2 \cdot \bar{u}_r^{cap} + \frac{1}{2} \cdot 1 \cdot u_s^{Ma} \\ &= \bar{u}_r^{cap} + \frac{u_s^{Ma}}{2} = \bar{u}_r^{cap} + \bar{u}_r^{Ma} \quad \checkmark \end{aligned} \quad (17)$$

7 Marangoni Flow Direction and Ring Suppression

7.1 Sign Convention

For suppression of the coffee ring, we need **inward surface flow**. This requires:

$$u_s^{Ma} < 0 \quad \Leftrightarrow \quad \frac{\partial \gamma}{\partial r} < 0 \quad (\text{surface tension decreases outward}) \quad (18)$$

7.2 Physical Mechanisms

Thermal Marangoni in Water

In water, $d\gamma/dT \approx -1.5 \times 10^{-4} \text{ N}/(\text{m}\cdot\text{K})$:

- Evaporation cools the surface, especially at the edge (higher J)
- Edge is colder $\Rightarrow$ edge has **higher** surface tension
- $\partial\gamma/\partial r > 0 \Rightarrow$ surface flow is **outward**
- **This enhances the coffee ring, not suppresses it!**

Solutal Marangoni for Suppression

To suppress the ring, we need $\partial\gamma/\partial r < 0$:

- Surfactants that accumulate at the edge (lower γ there)
- Solutes with positive $d\gamma/dc$ concentrated at the center
- Fluids where $d\gamma/dT > 0$ (rare)

7.3 Parameterization

We define a **Marangoni number** $Ma > 0$ to represent suppression (inward surface flow):

$$u_s^{Ma} = -Ma \cdot U_{cap} \cdot \frac{h}{h_0} \cdot g(\tilde{r}) \quad (19)$$

where:

- $U_{cap} = R/t_f$ is the characteristic capillary velocity
- $h_0 = R\theta_0/2$ is the initial apex height
- $g(\tilde{r}) = (1 - \tilde{r}^2 + \epsilon)^{-0.3}$ captures the edge enhancement
- The negative sign ensures inward flow when $Ma > 0$

8 Recirculation Physics

8.1 Why Recirculation Traps Particles

Consider the case of strong Marangoni flow ($Ma \gg 1$):

Layer	Capillary	Marangoni (inward)
Surface	+ (outward)	– (inward, strong)
Bulk	+ (outward)	– (inward, weak)
Total Surface	$u_s = \frac{11}{8}\bar{u}_r^{cap} - \frac{3}{4} u_s^{Ma} $	
Total Bulk	$u_b = \frac{5}{8}\bar{u}_r^{cap} - \frac{1}{4} u_s^{Ma} $	

For large enough Ma :

- $u_s < 0$ (surface flows inward)
- $u_b > 0$ or less negative (bulk flows outward or less inward)
- This creates a **recirculation cell**

8.2 Particle Trapping Mechanism

1. Particles near the edge are carried inward by surface flow
2. They descend to the bulk layer (via diffusion or vertical advection)
3. The bulk layer carries them back outward
4. They rise to the surface layer
5. The cycle repeats, **trapping particles** in the recirculation

This prevents particles from reaching the edge and depositing, **suppressing the coffee ring**.

9 Vertical Velocity and Exchange Rate

9.1 Vertical Velocity from Continuity

The incompressibility condition is:

$$\frac{1}{r} \frac{\partial(ru_r)}{\partial r} + \frac{\partial u_z}{\partial z} = 0 \quad (20)$$

For the Marangoni component:

$$u_z^{Ma}(r, z) = - \int_0^z \frac{1}{r} \frac{\partial}{\partial r} (ru_r^{Ma}) dz' = - \frac{z^2}{2rh} \frac{\partial}{\partial r} (ru_s^{Ma}) \quad (21)$$

At the midplane ($z = h/2$):

$$u_z^{Ma} \Big|_{z=h/2} = - \frac{h}{8r} \frac{\partial}{\partial r} (ru_s^{Ma}) \quad (22)$$

9.2 Exchange Rate Between Layers

Particles exchange between layers via two mechanisms:

1. **Diffusive exchange:**

$$k_{ex}^{diff} = \frac{D}{(h/4)^2} = \frac{16D}{h^2} \quad (23)$$

This represents diffusion across a distance $\sim h/4$ (layer center to interface).

2. **Advective exchange** (vertical Marangoni flow):

$$k_{ex}^{adv} = \frac{2|u_s^{Ma}|}{R} \quad (24)$$

Total Exchange Rate

$$k_{ex} = \frac{16D}{h^2} + \frac{2|u_s^{Ma}|}{R} \quad (25)$$

10 Two-Layer Transport Equations

10.1 Surface Layer Concentration

Surface Layer Equation

$$\frac{\partial c_s}{\partial t} + u_s \frac{\partial c_s}{\partial r} = \frac{D}{rh} \frac{\partial}{\partial r} \left(rh \frac{\partial c_s}{\partial r} \right) + \frac{c_s J}{\rho h} + k_{ex}(c_b - c_s) \quad (26)$$

Terms:

Term	Physical Meaning
$\partial c_s / \partial t$	Local accumulation in surface layer
$u_s \partial c_s / \partial r$	Advection by surface layer velocity
$\frac{D}{rh} \partial_r (rh \partial_r c_s)$	Radial diffusion
$c_s J / (\rho h)$	Concentration by evaporation
$k_{ex}(c_b - c_s)$	Exchange with bulk layer

10.2 Bulk Layer Concentration

Bulk Layer Equation

$$\frac{\partial c_b}{\partial t} + u_b \frac{\partial c_b}{\partial r} = \frac{D}{rh} \frac{\partial}{\partial r} \left(rh \frac{\partial c_b}{\partial r} \right) + k_{ex}(c_s - c_b) - \frac{2S_{dep}}{h} \quad (27)$$

Terms:

Term	Physical Meaning
$\partial c_b / \partial t$	Local accumulation in bulk layer
$u_b \partial c_b / \partial r$	Advection by bulk layer velocity
$\frac{D}{rh} \partial_r (rh \partial_r c_b)$	Radial diffusion
$k_{ex}(c_s - c_b)$	Exchange with surface layer
$2S_{dep}/h$	Deposition (only from bulk layer)

Note: Deposition only occurs from the bulk layer because particles must be near the substrate to deposit.

10.3 Deposition

The deposition model remains the same as the single-layer case, but depends on bulk concentration:

$$\frac{\partial \sigma}{\partial t} = S_{dep} = k_{dep} \cdot c_b \cdot \exp\left(-\frac{h}{h^*}\right) \quad (28)$$

11 Physical Regimes

The behavior depends on the Marangoni number:

Ma	Surface Flow	Bulk Flow	Result
0	Outward	Outward	Coffee ring
Low	Less outward	Outward	Weak suppression
Moderate	$\approx$ Zero	Outward	Partial suppression
High	Inward	Outward	Uniform/central deposit

The critical Marangoni number for zero surface velocity is approximately:

$$Ma_{crit} \sim \frac{11}{6} \cdot \frac{\bar{u}_r^{cap}}{U_{cap}} \cdot \frac{h_0}{h} \cdot \frac{1}{g(\tilde{r})} \quad (29)$$

12 Complete System of Equations

Complete Two-Layer System

Geometry and evaporation (same as single-layer):

$$h(r, t) = \frac{(R^2 - r^2)\theta(t)}{2R} \quad (30)$$

$$\theta(t) = \theta_0 \left(1 - \frac{t}{t_f}\right) \quad (31)$$

$$J(r) = J_0 \left(1 - \frac{r^2}{R^2}\right)^{-\lambda} \quad (32)$$

Velocities:

$$\bar{u}_r^{cap} = \frac{1}{rh} \int_0^r \left[\frac{h(r')}{t_f - t} - \frac{J(r')}{\rho} \right] r' dr' \quad (33)$$

$$u_s^{Ma} = -Ma \cdot U_{cap} \cdot \frac{h}{h_0} \cdot g(\tilde{r}) \quad (34)$$

$$u_s = \frac{11}{8} \bar{u}_r^{cap} + \frac{3}{4} u_s^{Ma} \quad (35)$$

$$u_b = \frac{5}{8} \bar{u}_r^{cap} + \frac{1}{4} u_s^{Ma} \quad (36)$$

Exchange rate:

$$k_{ex} = \frac{16D}{h^2} + \frac{2|u_s^{Ma}|}{R} \quad (37)$$

Transport equations:

$$\frac{\partial c_s}{\partial t} + u_s \frac{\partial c_s}{\partial r} = \frac{D}{rh} \frac{\partial}{\partial r} \left(rh \frac{\partial c_s}{\partial r} \right) + \frac{c_s J}{\rho h} + k_{ex}(c_b - c_s) \quad (38)$$

$$\frac{\partial c_b}{\partial t} + u_b \frac{\partial c_b}{\partial r} = \frac{D}{rh} \frac{\partial}{\partial r} \left(rh \frac{\partial c_b}{\partial r} \right) + k_{ex}(c_s - c_b) - \frac{2S_{dep}}{h} \quad (39)$$

Deposition:

$$\frac{\partial \sigma}{\partial t} = k_{dep} \cdot c_b \cdot \exp \left(-\frac{h}{h^*} \right) \quad (40)$$

13 Numerical Implementation

13.1 Discretization

The numerical scheme follows the single-layer model with extensions:

1. Compute $\bar{u}_r^{cap}$ using trapezoidal integration
2. Compute u_s^{Ma} from equation (19)
3. Compute layer velocities u_s and u_b
4. Compute exchange rate k_{ex}
5. Update c_s and c_b using upwind advection and central diffusion
6. Update deposition σ
7. Update height h

13.2 Upwind Scheme for Both Layers

Since u_s and u_b can have different signs:

- For surface layer: use upwind based on sign of u_s
- For bulk layer: use upwind based on sign of u_b

13.3 Stability

The exchange term adds a constraint:

$$\Delta t < \min \left(\frac{\Delta r}{\max |u_s|}, \frac{\Delta r}{\max |u_b|}, \frac{\Delta r^2}{2D}, \frac{1}{k_{ex,max}} \right) \quad (41)$$

14 Comparison: Single-Layer vs Two-Layer

Feature	Single-Layer	Two-Layer
Layers	1	2 (surface + bulk)
Variables	$c(r, t)$	$c_s(r, t), c_b(r, t)$
Velocity	$\bar{u}_r$	$u_s \neq u_b$
Marangoni	No	Yes
Recirculation	No	Yes
Ring suppression	No	Yes
Computational cost	Low	Moderate

15 Summary

The two-layer model captures Marangoni-driven recirculation by:

1. Deriving velocity profiles from lubrication theory with Marangoni stress boundary conditions
2. Splitting the film into surface and bulk layers with different average velocities
3. Tracking concentration in each layer separately with exchange between layers
4. Modeling deposition from the bulk layer only

Key results:

- Surface layer velocity: $u_s = \frac{11}{8}\bar{u}_r^{cap} + \frac{3}{4}u_s^{Ma}$
- Bulk layer velocity: $u_b = \frac{5}{8}\bar{u}_r^{cap} + \frac{1}{4}u_s^{Ma}$
- For $Ma > 0$: surface flows inward, creating recirculation
- Recirculation traps particles and suppresses the coffee ring

16 References

1. Deegan, R. D., et al. (1997). “Capillary flow as the cause of ring stains from dried liquid drops.” *Nature*, 389, 827-829.
2. Hu, H., & Larson, R. G. (2002). “Evaporation of a sessile droplet on a substrate.” *J. Phys. Chem. B*, 106, 1334-1344.
3. Hu, H., & Larson, R. G. (2005). “Analysis of the microfluid flow in an evaporating sessile droplet.” *Langmuir*, 21, 3963-3971.
4. Hu, H., & Larson, R. G. (2006). “Marangoni effect reverses coffee-ring depositions.” *J. Phys. Chem. B*, 110, 7090-7094.

5. Popov, Y. O. (2005). “Evaporative deposition patterns: Spatial dimensions of the deposit.” *Phys. Rev. E*, 71, 036313.
6. Ristenpart, W. D., et al. (2007). “Influence of substrate conductivity on circulation reversal in evaporating drops.” *Phys. Rev. Lett.*, 99, 234502.